

TRACE-v2 Equation Ledger

Implemented equations, verified status, and planned changes

Working document — generated for architecture tracking

July 31, 2026

Legend

- **Green** Implemented *and* numerically verified correct. No change planned.
- **Red** Implemented but *defective*, or manuscript text disagrees with code. Must change.
- **Blue** Planned replacement. Mathematically derived and (where stated) benchmarked.
- **Orange** Derived from VMK, not yet implemented. Candidate, not yet committed.

Manuscript equation numbers follow `docs/TRANSFORMERS_ACE_METHODS.tex`. VMK = Varshalovich, Moskalev & Kheronskii, *Quantum Theory of Angular Momentum* (World Scientific 1988; OA 2021), cited as VMK ch.sec(eq). Companion documents: `docs/PEER_REVIEW.md` (defects), `docs/ANGULAR_MOMENTUM_REFERENCE.md` (derivations).

1 Status summary

Block	Equations	Status	Action
Local energy decomposition	(2)	verified	—
Periodic edges	(4)–(6)	verified	—
Cutoff envelope	(7)–(8)	verified	—
Radial basis	(9)–(10)	verified	—
Edge tensor (ACE A-basis)	(11)–(12)	verified	P1: refactor, bit-exact
Clebsch–Gordan product	(13)	incomplete	D5 + P2
Neighbour density	(14)	verified	—
ACE correlations	(15)–(17)	verified	P2: symmetrise
Query / key	(18)–(19)	verified	—
Attention logits	(20)	code ≠ text	D2 + P7
Cutoff softmax + null channel	(21)	verified	—
Values, residual, layer scale	(22)–(23)	checkpoint	D1 + P3
Scalar FFN	(24)–(28)	verified	P4: rescale
Locality statement	(29)–(30)	verified	—
Readout & forces	(31)–(32)	verified	P6: analytic kernel
Stress	(33)–(36)	verified	—
Losses	(37)–(40)	verified	—
Sobolev regulariser	(41)	noise-dominated	D4 + P5
Error metrics	(42)–(44)	offset-contaminated	D3
Muon optimiser	(45)–(50)	verified	—

2 Implemented equations — verified correct

2.1 Geometry, cutoff, radial basis

Eq. (4)–(6) Periodic directed edges — verified: PBC supercell test 2×10^{-12} eV

$$\mathbf{r}_{ij} = \mathbf{r}_j - \mathbf{r}_i + \mathbf{S}_{ij}\mathbf{h}, \quad d_{ij} = \|\mathbf{r}_{ij}\|_2, \quad \hat{\mathbf{r}}_{ij} = \mathbf{r}_{ij}/d_{ij}$$
$$\mathcal{N}(i) = \{(j, \mathbf{S}_{ij}) : d_{ij} < r_c\}$$

Code: `model.py:784--788`. Sender/receiver/shift convention matches ASE `neighbor_list('ijS')` exactly. Note 754/979 CsPbI₃ frames violate the minimum-image condition, so one (i, j) pair legitimately yields several edge tokens — handled correctly.

Eq. (7)–(9) Quintic envelope and Bessel basis — verified C^2 ; $F \propto \varepsilon^3$ at r_c

$$f_c(d) = 1 - 10x^3 + 15x^4 - 6x^5 \equiv (1-x)^3(1+3x+6x^2), \quad x = d/r_c$$

$$B_n(d) = f_c(d) \sqrt{\frac{2}{r_c}} \frac{\sin(\omega_n d/r_c)}{d}, \quad \omega_n = n\pi$$

Normalisation $\int_0^{r_c} B_n^2 dd = 1$ correct; $d \rightarrow 0$ analytic limit $n\pi/r_c$ correct (`physics.py:52--56`). The factored envelope avoids float32 cancellation near $x = 1$.

2.2 ACE descriptor

Eq. (11)–(12) Edge tensor — verified rank-1 to 3.4×10^{-16} (VMK 5.6(14))

$$\mathbf{a}_{ij} = \text{TP}_{\text{dens}}\left(\mathbf{e}_j, \{Y_{\ell m}(\hat{\mathbf{r}}_{ij})\}_{\ell=0}^{\ell_{\max}}; \text{MLP}_r[\mathbf{B}(d_{ij})]\right), \quad a_{ij,cm}^{(\ell,(-1)^\ell)} = Y_{\ell m}(\hat{\mathbf{r}}_{ij}) \sum_q R_{cq}^{(\ell)}[\mathbf{B}(d_{ij})] e_{j,q}$$

VMK 5.6(14) states $\{Y_{\ell_1}(\Omega) \otimes Y_{\ell_2}(\Omega)\}_{LM} = \sqrt{\frac{(2\ell_1+1)(2\ell_2+1)}{4\pi(2L+1)}} C_{\ell_1 0 \ell_2 0}^{L0} Y_{LM}(\Omega)$. Since $\mathbf{e}_j$ is pure $0e$, the product collapses exactly to the component form above. Verified: the (c, m) matrix has rank 1 for every edge; recovered constant $= 1/\sqrt{2\ell+1}$ exactly.

Eq. (14)–(17) Density, correlations, centre state — permutation invariant

$$\mathbf{A}_i = \sum_{j \in \mathcal{N}(i)} \mathbf{a}_{ij}, \quad \mathbf{C}_i^{[1]} = \mathbf{A}_i, \quad \mathbf{C}_i^{[q+1]} = \text{TP}_{q+1}\left(\mathbf{C}_i^{[q]}, \mathbf{A}_i\right)$$

$$\mathbf{h}_i^{(0)} = \text{Center}(\mathbf{e}_i) + \sum_{q=1}^{q_{\max}} \text{Linear}_q\left(\mathbf{C}_i^{[q]}\right)$$

Matches VMK 3.1(21). `correlation_order` = $q_{\max} + 1 = 4$ confirmed. *Caveat retained in the manuscript and correct:* repeated neighbour indices mean $\mathbf{C}^{[q]}$ is not a pure $(q+1)$ -body term. VMK 3.2(22)–(23) makes this exact — for a single neighbour the whole hierarchy collapses to one $Y_{\ell m}$.

2.3 Attention

Eq. (21) Cutoff-weighted softmax with unit null channel — verified algebraically exact

$$\alpha_{ijp}^{(t)} = \frac{f_c(d_{ij}) \exp(\tilde{s}_{ijp}^{(t)})}{1 + \sum_{k \in \mathcal{N}(i)} f_c(d_{ik}) \exp(\tilde{s}_{ikp}^{(t)})}$$

`_segment_cutoff_softmax` reproduces this identically after the $\exp(\max)$ rescaling. The “1+” null token is what prevents normalisation from cancelling f_c in the single-neighbour limit — confirmed: E continuous and $F \propto \varepsilon^3$ as the last edge leaves r_c . **Note:** $\sum_j \alpha_{ij} < 1$; this is a *gated sum*, not a convex combination.

Eq. (31)–(36) Readout, forces, stress — verified against finite differences

$$E = \sum_i \text{MLP}_E(\bar{\mathbf{h}}_{i,\ell=0}) + E_{\text{ref}}, \quad \mathbf{F}_i = - \left. \frac{\partial E}{\partial \mathbf{r}_i} \right|_{\mathbf{h}}$$

$$\sigma_{aa} = \frac{1}{V} \left. \frac{\partial E}{\partial \eta_a} \right|_{\eta=0}, \quad \sigma_{xy} = \frac{1}{2V} \left. \frac{\partial E}{\partial \eta_4} \right|_{\eta=0}$$

Forces vs. central FD: 2.6×10^{-6} eV/Å. $\sum_i \mathbf{F}_i = 4 \times 10^{-17}$. Stress vs. ASE `numeric_stress`: 7×10^{-5} relative, including the factor $\frac{1}{2}$ on shear (ASE divides by 4dV). Deployment virial $\mathbf{W} = -V\boldsymbol{\sigma}$ consistent (`deploy.py:169`).

Eq. (37)–(40), (45)–(50) Losses and Muon — line-by-line match

All four loss/metric definitions match the trainer, including the stated convention that stress-free batch items contribute zero while the denominator stays $|\mathcal{B}|$. Muon reproduces VMK-independent reference exactly: momentum, Nesterov form, transpose rule, five quintic Newton–Schulz steps with $(a, b, c) = (3.4445, -4.7750, 2.0315)$, $\sqrt{\max(1, m/n)}$ scaling, decoupled decay $\Theta_{t+1} = (1 - \eta\lambda)\Theta_t - \eta U_t$.

3 Defects — implemented but wrong

D1 Eq. (23) Layer scale breaks $O(3)$ equivariance in the published checkpoint

$$\tilde{\mathbf{h}}_i^{(t)} = \mathbf{h}_i^{(t)} + \gamma_{\text{attn}}^{(t)} \odot \mathbf{u}_i^{(t)}$$

The manuscript states γ_{attn} is *shared over all magnetic components m within each irrep copy*. The parameter is stored *per component* (`irreps.dim=240`). The averaging fix (`_irrepwise_layer_scale`) landed 2026-07-11; `training/model.pt` was written 2026-06-28.

Proof the checkpoint predates the fix: with the fixed code all components of a copy receive identical gradients from a uniform init (0.01), so they can never diverge. Measured within-copy spread: 1.157 ($\ell=1$), 1.356 ($\ell=2$).

Magnitude: energy spread under random rotation = 5.96 meV/atom (max 7.31); force mismatch 0.115 eV/Å. Compare reported accuracy 2.44 meV/atom, 0.0388 eV/Å. **The symmetry violation is 2–3× larger than the model’s own error.** Same checkpoint through the fixed path differs by 3.59 meV/atom and 0.011 eV/Å RMS. **Requires retraining and regeneration of every figure.**

D2 Eq. (20) Attention temperature: code $\neq$ manuscript, and train $\neq$ inference

$$s_{ijp}^{(t)} = \frac{(\mathbf{q}_{ip}^{(t)})^\top \mathbf{k}_{ijp}^{(t)}}{\sqrt{d_k}} + b_p^{(t)}(\mathbf{B}(d_{ij})) - \text{softplus}(\lambda_p^{(t)}) d_{ij}, \quad \hat{s}_{ijp}^{(t)} = s_{ijp}^{(t)} / T_{\text{att}}(\tau)$$

The text defines $T_{\text{att}}(\tau)$ as a function of *epoch* τ . The code (`train.py:694--706`) additionally multiplies by a feedback factor $(\mathcal{L}_F/0.5)^{-1/4}$ driven by the running force-loss EMA:

$$T_{\text{att}} = T_{\text{base}}(\tau) \cdot (\text{EMA}[\mathcal{L}_F]/T_{\text{ref}})^\gamma, \quad T_{\text{ref}} = 0.5, \quad \gamma = -0.25.$$

At the final force MSE 1.39×10^{-3} this gives $T_{\text{att}} \approx 4.36$, while validation (`train.py:906`) and the ASE calculator both use $T_{\text{att}} = 1$. Measured on the trained checkpoint: E error $2.374 \rightarrow 5.156$ meV/atom between the two temperatures. **A different function is being trained than is deployed.**

D3 Eq. (42) Reported energy RMSE dominated by a drifting offset

$$\text{RMSE}_E = \left[\frac{\sum_s N_s (\Delta E_s / N_s)^2}{\sum_s N_s} \right]^{1/2}$$

The formula is correct; the *reference* is a single scalar shift (−31.8566 eV/atom, `solve_atomic_energies: false`). Decomposition: epoch 76 $\rightarrow$ mean signed error +1.832, s.d. 1.509 (RMSE 2.374); epoch 100 $\rightarrow$ mean +0.176, s.d. 1.521 (RMSE 1.531). **The offset-free error is identical (1.51) — the reported RMSE differs by 55% purely from a wandering energy zero.** This explains the epoch-to-epoch oscillation (5.81 $\rightarrow$ 2.11 $\rightarrow$ 2.13 $\rightarrow$ 4.20) while force RMSE stays flat, and the non-reproducibility across e3nn versions.

D4 Eq. (41) Sobolev term measures dropout variance, not curvature

$$\mathcal{L}_{\text{Sob}} = \frac{1}{|\mathcal{B}|} \sum_s \left[E_s(\mathbf{r}_s + \boldsymbol{\delta}_s) - E_s(\mathbf{r}_s) + \sum_i \mathbf{F}_{si} \cdot \boldsymbol{\delta}_{si} \right]^2$$

The two energies are evaluated in *separate forward passes* with *independent* attention-dropout masks ($p = 0.03$, active in `model.train()`). Measured on a 40-atom cell: dropout-induced s.d.(E) = 0.0504 eV versus an intended residual of 0.0389 eV — **noise/signal = 1.3.**

D5 Eq. (13) Missing channel sums

$$[\mathbf{x} \otimes_{\text{CG}} \mathbf{y}]_{cm}^{(\ell,p)} = \sum_{\substack{\ell_1 p_1 c_1 m_1 \\ \ell_2 p_2 c_2 m_2}} W_{c_1 c_2 c}^{\ell_1 \ell_2 \ell} C_{\ell_1 m_1, \ell_2 m_2}^{\ell m} x_{c_1 m_1}^{(\ell_1, p_1)} y_{c_2 m_2}^{(\ell_2, p_2)}$$

As printed the channel indices c_1, c_2 are not summed. Compare VMK 3.1(20). Editorial only — the code is correct.

4 Planned changes

P1 Species-factorised edge features — bit-exact, 1.61× faster, removes 41% of parameters

Current (`physics.py:425`): a per-edge weight tensor of 1792 floats is materialised, then contracted with the 64-dim species embedding.

Derivation. By VMK 5.6(14) the output depends on the final radial layer W_2 and the species embedding E *only* through

$$M[s, h, c] = \sum_q W_2[h, q, c] E[s, q], \quad a_{e,c,m}^{(\ell)} = \frac{1}{\sqrt{2\ell+1}} \left(\sum_h \text{hidden}_h(d_e) M[s(e), h, c] \right) Y_{\ell m}(\hat{r}_e).$$

M has only $n_{\text{spec}} \times 32 \times 28$ degrees of freedom.

n_{spec}	W_2 + used E	d.o.f. in M	redundant
2 (water; C/H reaction)	57,472	1,792	96.9%
3 (CsPbI ₃)	57,536	2,688	95.3%
8	57,856	7,168	87.6%

Verified. A $50\times$ perturbation of W_2 inside the null space of $W_2 \rightarrow M$ changes the output by 9.2×10^{-13} (output scale 1.96). For CsPbI₃: 54,656 parameters = 41% of the 132,005-parameter model cannot affect any prediction.

Benchmarked. Factorised path reproduces the current output to 5.7×10^{-16} relative; $3.395 \rightarrow 2.107$ ms (1.61×); dominant matmul $34.9 \rightarrow 0.72$ MFLOP (49×); intermediate $[608, 1792] \rightarrow [608, 32]$ (57× smaller).

Two options. (a) Exact refactor — keep W_2 , precompute M once per species per forward. Identical model, identical parameters, 1.61× faster. (b) Re-parameterise — learn M directly (2,688 params). *Same function class*, but 41% fewer parameters, better conditioning, and no wasted Muon/AdamW update budget or weight decay on null directions. **Recommend (b).**

P2 Symmetrise the first ACE contraction — removes 4,136 dead weights

Current: `contractions[0]` (`density, density`) is $\text{TP}(\mathbf{A}_i, \mathbf{A}_i)$ — the *same* tensor in both slots.

VMK 3.1(27) (commuting tensors): $\{\mathfrak{M}_{J_1} \otimes \mathfrak{M}_{J_2}\}_{JM} = (-1)^{J_1+J_2-J} \{\mathfrak{M}_{J_2} \otimes \mathfrak{M}_{J_1}\}_{JM}$.

Consequently, for $\text{TP}(\mathbf{A}_i, \mathbf{A}_i)$:

Redundancy source	Dead weights
Antisymmetric part of $W[c_1, c_2, c]$ on all $\ell_1 = \ell_2$ paths	2,600
Mirror pairs $(\ell_1, \ell_2) \rightarrow \ell$ vs $(\ell_2, \ell_1) \rightarrow \ell$	1,536
Total	4,136 of 8,768 = 47.2%

Verified: a large purely antisymmetric perturbation changes $\text{TP}(\mathbf{A}_i, \mathbf{A}_i)$ by 2×10^{-14} ; mirror-pair sensitivities collinear to 6×10^{-16} .

Change: parameterise only the symmetric part on $\ell_1 = \ell_2$ paths and merge mirror pairs (equivalently use `o3.TensorSquare`). Frees ~4k parameters at *identical* expressivity. Alternative with *more* expressivity: contract $\mathbf{A}_i$ against a distinct learned linear image of $\mathbf{A}_i$, restoring the antisymmetric paths as genuine degrees of freedom.

P3 Re-shape the layer scale so the broken state is unrepresentable

Store γ_{attn} with *one scalar per irrep copy* (shape = $\sum_{\ell} \text{mul}_{\ell}$) instead of one per component ($= \sum_{\ell} \text{mul}_{\ell}(2\ell+1)$), rather than masking a 240-vector at forward time:

$$\gamma \in \mathbb{R}^{64+32+16=112} \quad \text{replacing} \quad \gamma \in \mathbb{R}^{240}.$$

Add a CI test asserting rotational invariance of a *loaded checkpoint*. The existing test hand-sets the parameter and exercises the fixed path, so it cannot catch D1.

P7 Make T_{att} consistent between training, validation and deployment

Either set `temperature_force_exponent`: 0 so $T_{\text{att}} \rightarrow 1$ before training ends, or pass the same `temperature_scale` into the validation forward. Then correct Eq. (20) to describe whichever is used. **Recommend annealing to 1**, so the deployed potential is the one that was optimised.

5 Derived from VMK — candidates, not yet committed

P4 Rescale the FFN invariants by $(2\ell + 1)$

$$n_{ic\ell}^{(t)} = \sum_{m=-\ell}^{\ell} |\tilde{h}_{icm}^{(t,\ell)}|^2 \quad \longrightarrow \quad \hat{n}_{ic\ell}^{(t)} = \frac{1}{2\ell + 1} \sum_m |\tilde{h}_{icm}^{(t,\ell)}|^2$$

By VMK 3.1(35), $(\mathfrak{M}_J \cdot \mathfrak{M}_J) = (-1)^{-J} \sqrt{2J+1} \{\mathfrak{M}_J \otimes \mathfrak{M}_J\}_{00}$, so the invariants carry an ℓ -dependent natural scale ($\sqrt{3}$ at $\ell=1$, $\sqrt{5}$ at $\ell=2$). **LayerNorm** absorbs this at inference but it biases initialisation and early optimisation. Cost: zero.

P5 Exact irreducible Sobolev term (replaces the stochastic finite difference of D4)

$$(\mathbf{u} \cdot \nabla)^n = (-1)^n \sum_{\ell_2 \dots \ell_n} (-1)^{\ell_n} \left(\{ \dots \{ \mathbf{u}_1 \otimes \mathbf{u}_1 \}_{\ell_2} \dots \otimes \mathbf{u}_1 \}_{\ell_n} \cdot \{ \dots \{ \nabla_1 \otimes \nabla_1 \}_{\ell_2} \dots \otimes \nabla_1 \}_{\ell_n} \right) \quad \text{VMK 3.2(44)}$$

The $n = 2$ term is exactly the Hessian contraction $\mathcal{L}_{\text{Sob}}$ is trying to penalise, decomposed into $\ell = 0$ and $\ell = 2$ pieces. Using the closed form removes the two-forward-pass structure and therefore the dropout contamination of D4 entirely, and makes the regulariser deterministic.

P6 Analytic force kernel for the AOT/LAMMPS path

$$\nabla[f(r)Y_{\ell m}] = -\sqrt{\frac{\ell+1}{2\ell+1}} \left(\frac{df}{dr} - \frac{\ell}{r} f \right) \mathbf{Y}_{\ell m}^{\ell+1} + \sqrt{\frac{\ell}{2\ell+1}} \left(\frac{df}{dr} + \frac{\ell+1}{r} f \right) \mathbf{Y}_{\ell m}^{\ell-1} \quad \text{VMK 5.8(9)}$$

Closed irreducible form for the gradient of one edge feature. Enables an analytic (non-autograd) force kernel for the compiled LAMMPS path, and an independent check on the autograd forces. Requires VMK Ch. 7.3 (vector spherical harmonics) first.

P8 Multipole expansion, if long-range electrostatics is added

$$\frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|} = \frac{4\pi}{r_2} \sum_{\ell} \frac{1}{2\ell + 1} \left(\frac{r_1}{r_2} \right)^{\ell} (\mathbf{Y}_{\ell}(\Omega_1) \cdot \mathbf{Y}_{\ell}(\Omega_2)), \quad r_1 < r_2 \quad \text{VMK 5.17(21)}$$

Composes directly with the existing ℓ -channels. §IV names the absence of a long-range term as a scope limitation; this is the form that would not require separate Ewald machinery.

6 Structural results worth stating in the manuscript

S1 Natural parity is the Gaunt selection rule

$C_{\ell_1 \ell_2 0}^{L0} = 0$ unless $\ell_1 + \ell_2 + L$ is even (VMK 5.6(9)). Verified in e3nn's real basis: every odd combination ($1, 1 \rightarrow 1$; $1, 2 \rightarrow 2$; $2, 2 \rightarrow 1$; $2, 2 \rightarrow 3$) evaluates to *identically zero*. So the $(\ell, (-1)^{\ell})$ truncation is not an arbitrary economy — it keeps precisely the couplings that survive for harmonics of a single direction. One sentence in §II A.

S2 Association order is a $6j/9j$ basis change, *not* extra expressivity

VMK 3.3(1) relates left- and right-associated triple products by a $6j$ symbol; VMK 3.3(11) relates the pairwise scheme $\{\{P \otimes Q\} \otimes \{R \otimes S\}\}$ to the left-associated one by a $9j$. At fixed ℓ_{max} with fully connected channel mixing, all association orders reach the same tensor space. **Any advantage of the v4 cumulant coupling must therefore be claimed as conditioning/optimisation, not capacity.**

S3 Higher q_{max} requires ≥ 2 distinct neighbours

VMK 3.2(22)–(23): a chain of tensor products of a *single* vector equals one spherical harmonic times a power of the length. Hence for a one-neighbour environment the entire hierarchy $\mathcal{C}^{[1]} \dots \mathcal{C}^{[q_{\text{max}}]}$ carries no more information than $\{|\mathbf{r}|, Y_{\ell m}(\hat{\mathbf{r}})\}$. Testable prediction: accuracy gain from **correlation_order** should scale with mean neighbour count — weaker for gas-phase methyl migration than for dense CsPbI₃.

7 Negative results — do not re-try

Hypothesis	Measured	Verdict
Hand-write tp.density keeping per-edge weights	2.281 $\rightarrow$ 2.237 ms	1.02 $\times$; the TP is not the bottleneck, the per-edge weight <i>tensor</i> is
Fuse the H per-head o3.Linear value projections	$H=2$: 1.13 $\times$; $H=4$: 1.01 $\times$; $H=8$: 0.98 $\times$	not worth it; worse at large H
Addition theorem (VMK 5.17(9)) to replace $\sum_m$ with Legendre polynomials	cost $n(\ell+1)^2$ vs $n^2\ell$	wins only when $\ell_{\text{max}} \gtrsim \bar{n}$; a loss at $\bar{n} \approx 15$, $\ell_{\text{max}} = 2$

Profile for reference (40-atom cell, 608 edges, 1 CPU thread, energy+forces = 20.8 ms): **aten::bmm** 38.4% self CPU (1440 calls), **aten::einsum** 21.4% of total (600 calls), **aten::fill_** 6.4%, **aten::mm** 5.1%. The batched-matmul time is dominated by the per-edge weight contraction that P1 removes.

Verification scripts and raw numbers: see `docs/PEER_REVIEW.md` and `docs/ANGULAR_MOMENTUM_REFERENCE.md`. All figures in this ledger were measured on the shipped `training/model.pt` (epoch 76) and `training/train.extxyz` (979 frames).